

Related Manuscripts and Preprint Disclosure

Soroush Vahidi — COAP submission, 12 June 2026

Current submission. *IPSNS for Minimum Weighted Feedback Arc Set on Sparse Digraphs* (Computational Optimization and Applications).

Public preprint. Vahidi, S.; Koutis, I. “Minimum Weighted Feedback Arc Sets for Ranking from Pairwise Comparisons.” arXiv:2412.16181 (December 2024). **Status:** public preprint; not a journal submission. **Overlap:** MWFAS problem framing, early heuristic ingredients (local-ratio peeling, feasible reinsertion, topological extraction). **COAP increment:** IPSNS-first unified framework; formal analysis; expanded sparse benchmarks; exact and MIP validation; LOLIB scope boundary; holdout; EXP10/EXP11; Online Resource 1.

LR-TA predecessor manuscript. “Fast Local-Ratio Cycle Reduction with Topological Add-Back for Weighted Feedback Arc Sets.” **Status:** earlier unpublished author manuscript; no longer under consideration. **Overlap:** LR-TA Phase I/II at code level. **COAP increment:** IPSNS integration, dual-seed framework, broader experiments, formal properties, EXP10/EXP11.

IPSNS predecessor manuscript. “Incumbent-Protected SCC-Neighborhood Search for the Weighted Feedback Arc Set Problem.” **Status:** earlier unpublished author manuscript; no longer under consideration. **Overlap:** IPSNS core and WMSF-style seed on smaller benchmark suite. **COAP increment:** LR-TA seed integration, unified exposition, formal analysis, expanded validation program.

Merged framework predecessor manuscript. “A Reproducible Local-Ratio and SCC-Refinement Framework for Weighted Ordering in Directed Graphs.” **Status:** earlier unpublished author manuscript; no longer under consideration. **Overlap:** merged LR-TA/WMSF/IPSNS experimental narrative. **COAP increment:** Springer COAP template, formal analysis, holdout, EXP10, EXP11, refined scope language.

Pre-holdout predecessor manuscript. “Local-Ratio Seeding and SCC-Based Refinement for the Minimum Weighted Feedback Arc Set Problem.” **Status:** earlier unpublished author manuscript; no longer under consideration. **Overlap:** near-identical scientific core pre-holdout/EXP10. **COAP increment:** holdout, EXP10 stochastic robustness, EXP11 extraction calibration, COAP-oriented presentation.

What is new in the COAP submission. IPSNS as the primary integrated contribution within a three-component framework; consolidated formal supporting analysis; exact and mixed-integer validation; 105-instance sparse study with external baselines; ablation, sensitivity, holdout, LOLIB transfer, EXP10, and EXP11; reproducible Online Resource 1; explicit distinction between removed-set and ordering objectives.

Summary table.

Prior item	Status	Shared with COAP	Current consideration
arXiv:2412.16181	Public preprint	MWFAS framing; LR-TA ingredients	N/A (preprint)
LR-TA predecessor	Inactive	LR-TA Phase I/II code	Not under consideration
IPSNS predecessor	Inactive	IPSNS core; WMSF seed	Not under consideration
Merged framework	Inactive	Merged experimental narrative	Not under consideration
Pre-holdout version	Inactive	Pre-holdout scientific core	Not under consideration

Note. This statement is provided as an editor-only supplementary document. It describes component-level overlap and the incremental contributions of the present COAP manuscript relative to earlier unpublished author work and the public preprint. No substantially overlapping manuscript is currently under consideration elsewhere.